

Ivan Torriani

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EDUCATION

California Polytechnic State University, San Luis Obispo

San Luis Obispo, CA

B.S. Computer Science (AI/ML Concentration)

May 2028

- **Achievements:** Dean's List; TSMG Scholarship Recipient; RCSA Scholarship Recipient; CTW Engineering Scholarship Winner (\$5,000, 2nd place out of 10+ teams)

EXPERIENCE

Director of Research (Incoming President)

May 2026 – Present

Cal Poly Computer Science & Artificial Intelligence (CSAI)

San Luis Obispo, CA

- Selected as incoming President of a 1,000+ member organization, the largest computing club at Cal Poly, after directing research operations for the 2026–2027 school year.
- Directed a team of associate directors and project leads across the club's research portfolio, defining project scope and success criteria to keep multi-team efforts on schedule.
- Spearheaded the launch of the CSAI Journal of Computer Science and a \$1,000 scholarship hackathon competition, both built to strengthen the research environment at Cal Poly.
- Delivered technical presentations and research showcases to audiences of 100+ students, faculty, and industry guests, driving recruitment and cross-team collaboration.

AI/ML Research Team Lead

Jan. 2026 – May 2026

Autonomous Vehicle Navigation with Meta Learning for Out-Of-Domain Terrain

San Luis Obispo, CA

- Owned product vision and strategy for a 4-phase autonomous navigation system after recognizing the market opportunity for self-driving software in non-urban scenes (e.g., disaster zones and rural areas) where standard models fail.
- Iterated through 4 versions of the unknown-object-detection feature using Claude Code by running experiments, qualitatively analyzing data, forming hypotheses on each, and running controlled tests to validate them, resulting in a 93% accuracy pipeline.
- Monitored and reviewed 4 teams within an Agile framework (1 hardware, 3 software) by defining weekly success criteria and running in-person meetings to confirm key results were met.

Software Engineering Intern

Sep. 2024 – Aug. 2025

Surfrider Foundation

San Luis Obispo, CA

- Shipped an interactive Shared Socioeconomic Pathway (SSP) climate calculator with React/Next.js to track the environmental footprints of company endeavors.
- Designed backend logic in TypeScript with MongoDB, including 8 REST APIs and cloud endpoints for real-time SSP projection.
- Deployed to production and monitored adoption metrics, growing to 100+ monthly visitors across Surfrider's coastal advocacy network.

AI/ML Research Intern

Sep. 2025 – May 2026

Cal Poly Computer Science Department

San Luis Obispo, CA

- Trained and optimized 10+ YOLO object-detection models on Google Cloud GPUs with a 1,000+ image dataset, eventually optimizing to a 30% mAP improvement via confusion-matrix and precision-recall data analysis.
- Integrated MiDaS monocular depth-estimation to produce real-time depth heatmaps and background blur, creating a \$5k–\$15k cheaper alternative to current LiDAR features.
- Boosted a Jetson Nano OpenCV pipeline running 2 models from 2 to 15 FPS (7.5× speedup) per stakeholder request by creating custom metric-distance algorithms.

AI/ML Research Intern

Sep. 2024 – Jun. 2025

Cal Poly Computer Science Department

San Luis Obispo, CA

- Built automated web-mining pipeline with cron to extract structured data from university pages with 400+ clubs and 2,000+ faculty records, eliminating ~3–4 hours of daily manual data collection per scrape.
- Developed vector-embedding indexing across 10,000+ records using cosine similarity search; benchmarked retrieval accuracy to validate embedding quality for downstream NLP tasks.
- Cut LLM API costs by 50% by refactoring sequential calls into asynchronous batch processing, reducing average request overhead across hundreds of nightly pipeline runs.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, C#, TypeScript, SQL

Frameworks: SciPy, PyTorch, TensorFlow, TensorRT, CUDA, React, Next.js, Node.js

Developer Tools: Git, Google Colab, Jupyter Notebook, Anaconda, MongoDB